

AYUSH NARAYAN

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EDUCATION

B.M.S. College of Engineering

Bachelor of Engineering in Information Science and Engineering | CGPA: 8.5/10

Bengaluru, Karnataka

Oct 2023 – Sep 2027

PROJECTS & EXPERIENCE

Automated Resume Analyzer – Self | Python, scikit-learn, TF-IDF, Logistic Regression, LinearSVC, NLTK, Gensim

- Designed an end-to-end supervised NLP pipeline for binary classification of resume text using sparse TF-IDF vector representations (`max_features = 5000`) to capture high-signal lexical features.
- Built a text preprocessing pipeline with tokenization (Gensim), stopwords removal (NLTK), normalization, and noise reduction; trained L2-regularized Logistic Regression and LinearSVC models on stratified train-test splits.
- Evaluated model generalization using accuracy, precision, recall, F1-score, and confusion matrices; integrated the complete pipeline into a real-time inference workflow for interactive prediction.

GyroCursor – Wireless Motion-Controlled Mouse – Self | Next.js, React, Kotlin, Python, FastAPI, PyAutoGUI

- Developed a full-stack wireless mouse control system that converts smartphone orientation data into real-time PC cursor movement using secure WebSocket (WSS) communication between an Android client and a Python backend.
- Built the Android client in Kotlin using rotation-vector sensors and converted sensor output into DeviceOrientation-compatible alpha, beta, and gamma angles; implemented configurable 10–100 Hz sampling, sensitivity control, and one-tap calibration.
- Implemented a FastAPI and WebSocket backend to process motion packets and generate host-level mouse movement and click events using PyAutoGUI, secured with self-signed TLS certificates for local-network communication.
- Developed and deployed a responsive dashboard using JavaScript, HTML, Next.js, React, and Tailwind CSS, featuring QR-based device pairing, real-time telemetry, and WebSocket ping/pong latency tracking.

AI-Powered Academic ERP Assistant (Ongoing) | LLM, RAG, REST APIs, Python

Industry Collaboration – Modelicon Infotech LLP | FRPS Funded Research (Rs. 4.5L)

- Contributing to the backend development of an AI-driven conversational ERP assistant as part of an industry-collaborative research initiative with **Modelicon Infotech LLP**, funded under **FRPS** (Rs. 4.5L grant).
- Designing a secure API integration between an LLM-based assistant and the institutional ERP system; developing a RAG pipeline to ground responses in structured ERP records and reduce hallucination.
- Building modular backend services for query parsing, authentication handling, and ERP request routing; supporting integration, testing, and scalability planning for campus-wide deployment.

Real-Time Network Congestion Detection & Mitigation | Python, Computer Networks, UDP, TCP, scikit-learn, iPerf3

- Developed a real-time network congestion monitoring system that probes network paths using ICMP, UDP, and TCP, collecting round-trip time (RTT) measurements with protocol fallback for continuous latency monitoring.
- Implemented RTT-based congestion analysis using moving-average smoothing and statistical features, including mean RTT, standard deviation, and latency elevation, to characterize network behavior.
- Applied an RBF-kernel Support Vector Machine (SVM) to classify network conditions as normal or congested, with an additional latency-trend predictor for early congestion detection.
- Designed a closed-loop congestion mitigation mechanism that dynamically adjusts probe frequency and traffic rate based on network state, and measured throughput with iPerf3 while logging RTT, congestion state, and throughput for performance analysis.

TECHNICAL SKILLS

Languages: C++, Python, Java, JavaScript (ES6), Kotlin

Frontend: React.js, Next.js, Tailwind CSS, HTML5, CSS3

Backend & APIs: FastAPI, Node.js, Express.js, REST APIs (Application Programming Interfaces), WebSockets, JSON

Databases: MySQL, Firestore, SQL, NoSQL, Database Design, Schema Design

Cloud & DevOps: Google Cloud, Docker, Kubernetes, Terraform, CI/CD (Continuous Integration/Continuous Deployment), Git, Cloud Infrastructure

Systems, Networking & Core CS: Linux/UNIX, Shell Scripting, TCP/IP, HTTP/HTTPS, Computer Networks, Data Structures and Algorithms, Object-Oriented Programming (OOP), Operating Systems, Database Management Systems (DBMS), Distributed Systems Fundamentals

Data & ML: NumPy, Pandas, scikit-learn, PyTorch, TF-IDF (Term Frequency-Inverse Document Frequency), NLP (Natural Language Processing), LLM (Large Language Model), RAG (Retrieval-Augmented Generation), Matplotlib

LEADERSHIP, RESEARCH & ACTIVITIES

Research Paper Presentation – ICNEXT-2026: Presented a research paper titled *Real-Time Congestion Detection and Mitigation Using Lightweight Probes and Dual SVM Classification* at the Next-Gen Engineering & Technology Conference. [Certificate]

Hackathon Team Lead: Led a 3-member team through 24–48 hour hackathons, overseeing solution architecture, task allocation, technical decision-making, and rapid prototype development.

Junior Coordinator – Information & Web Team, Phase Shift 2024: Assisted in managing and updating web content for the college's annual technical fest; coordinated with cross-functional teams and maintained clear documentation of content updates to ensure smooth and timely deployment.

Google Cloud Skill Badges: Completed hands-on Google Cloud skill badges covering cloud infrastructure, networking, Kubernetes Engine, Terraform-based infrastructure as code, and load balancing. [Certificates]